Machine Learning

COMS3007A/COMS3024A

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Formula Sheet

These formulas are provided for your reference during the examination.

Probability & Naïve Bayes

General Probability

- Sum Rule: $P(X) = \sum_Y P(X, Y)$
- Product Rule: $P(X, Y) = P(X|Y)P(Y)$
- Bayes' Rule: $P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$

Naïve Bayes Classifier

- Posterior (with conditional independence):
  $P(C = c_k|X = \mathbf{x}) = \frac{P(C=c_k)\prod_{i=1}^{n} P(X_i=x_i|C=c_k)}{P(X=\mathbf{x})}$
- MAP Solution: $c^* = \arg\max_{c_k} P(C = c_k|X = \mathbf{x})$
- Laplace Smoothing (k is smoothing parameter, V is number of possible values for feature X_i):
  $P(X_i = x_{ij}|C = c) = \frac{\text{count}(X_i=x_{ij},C=c)+k}{\text{count}(C=c)+kV}$

Decision Trees

- Entropy: $H(S) = -\sum_{i=1}^{C} p_i \log_2(p_i)$
- Information Gain:
  $Gain(S, A) = H(S) - \sum_{v\in Values(A)} \frac{|S_v|}{|S|} H(S_v)$

Linear Regression

- Model: $h_\theta(\mathbf{x}) = \sum_{j=0}^{d} \theta_j \phi_j(\mathbf{x}) = \theta^T \phi(\mathbf{x})$
  (with $\phi_0(\mathbf{x}) = 1$)
- Cost (SSE, single example i): $J_i(\theta) = \frac{1}{2}(h_\theta(\mathbf{x}^{(i)}) - y^{(i)})^2$
- Gradient Component (for θ_j, single example i):
  $\frac{\partial J_i(\theta)}{\partial \theta_j} = (h_\theta(\mathbf{x}^{(i)}) - y^{(i)})\phi_j(\mathbf{x}^{(i)})$
- Gradient Descent Update (single example i):
  $\theta_j \leftarrow \theta_j - \alpha(h_\theta(\mathbf{x}^{(i)}) - y^{(i)})\phi_j(\mathbf{x}^{(i)})$

Logistic Regression

- Sigmoid (Logistic) Function: $\sigma(z) = \frac{1}{1+e^{-z}}$
- Model: $h_\theta(\mathbf{x}) = \sigma(\theta^T \phi(\mathbf{x}))$
  (with $\phi_0(\mathbf{x}) = 1$)
- Cost (Cross-Entropy, single example i, $y^{(i)} \in \{0, 1\}$):
  $J_i(\theta) = -[y^{(i)} \log(h_\theta(\mathbf{x}^{(i)})) + (1 - y^{(i)}) \log(1 - h_\theta(\mathbf{x}^{(i)}))]$
- Gradient Component (for θ_j, single example i):
  $\frac{\partial J_i(\theta)}{\partial \theta_j} = (h_\theta(\mathbf{x}^{(i)}) - y^{(i)})\phi_j(\mathbf{x}^{(i)})$
- Gradient Descent Update (single example i):
  $\theta_j \leftarrow \theta_j - \alpha(h_\theta(\mathbf{x}^{(i)}) - y^{(i)})\phi_j(\mathbf{x}^{(i)})$

Neural Networks

- Sigmoid Activation: $g(z) = \sigma(z) = \frac{1}{1+e^{-z}}$
- Derivative of Sigmoid: $g'(z) = g(z)(1 - g(z))$
- Net input to neuron i in layer l: $z_i^{(l)} = \sum_j \theta_{ij}^{(l-1)} a_j^{(l-1)}$ (includes bias $a_0^{(l-1)} = 1$)
- Activation of neuron i in layer l: $a_i^{(l)} = g(z_i^{(l)})$
- Error Term (Output Layer L, neuron k): $\delta_k^{(L)} = a_k^{(L)} - y_k$
- Error Term (Hidden Layer l, neuron j):
  $\delta_j^{(l)} = \left(\sum_k \theta_{kj}^{(l)} \delta_k^{(l+1)}\right) g'(z_j^{(l)})$
- Gradient Component (for $\theta_{ij}^{(l-1)}$ for one example):
  $\frac{\partial J(\Theta)}{\partial \theta_{ij}^{(l-1)}} = a_j^{(l-1)} \delta_i^{(l)}$
- Weight Update: $\theta_{ij}^{(l-1)} \leftarrow \theta_{ij}^{(l-1)} - \alpha \frac{\partial J(\Theta)}{\partial \theta_{ij}^{(l-1)}}$

K-means Clustering

- Euclidean Distance (squared):
  $d(\mathbf{x}_i, \mathbf{x}_j)^2 = ||\mathbf{x}_i - \mathbf{x}_j||^2 = \sum_{k=1}^{D}(x_{i,k} - x_{j,k})^2$ (for D dimensions)
- K-means Objective (sum of squared errors):
  $J = \sum_{i=1}^{N} ||\mathbf{x}_i - \mu_{c_i}||^2$ (μ_{c_i} is centroid of cluster for $\mathbf{x}_i$)
- Centroid Update for cluster k: $\mu_k = \frac{1}{|S_k|} \sum_{\mathbf{x}_i \in S_k} \mathbf{x}_i$

Evaluation Metrics (Classification)

(TP: True Positives, TN: True Negatives, FP: False Positives, FN: False Negatives)

- Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$
- Precision (positive class): $\frac{TP}{TP+FP}$
- Recall / TPR (positive class): $\frac{TP}{TP+FN}$
- F1-Score (positive class): $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Reinforcement Learning

- Discount Factor: $0 \leq \gamma \leq 1$
- State-Value Function:
  $V^{\pi}(s) = E_{\pi}[R_{t+1} + \gamma V^{\pi}(S_{t+1})|S_t = s]$
- TD Learning Update for $V(S_t)$:
  $V(S_t) \leftarrow V(S_t) + \alpha[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)]$